



# Flag Maker

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# Flag Maker Activity

- You'll be pretending to be the computer drawing the flag of **Mauritius** (on the right).
- You'll use paper and markers
- Follow how a computer might draw it as if each box were an element in an array and the computer is iterating through the array in order.
- You must fill in the squares and **color one square at a time** (do not color in multiple squares at one time).



Img from: <https://en.wikipedia.org/wiki/Mauritius>

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 |
| 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 |

# Flag Maker Activity

You must color one square at a time.

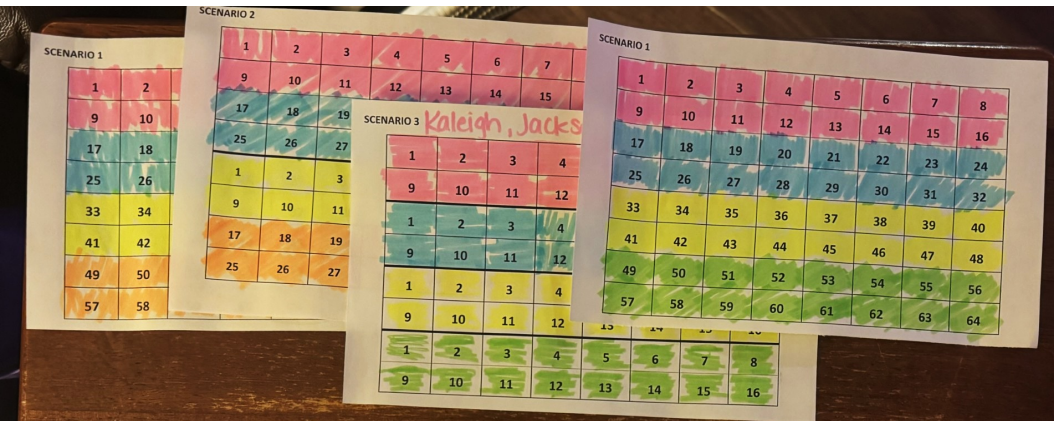
Do not color in multiple squares at one time

**CORRECT!**



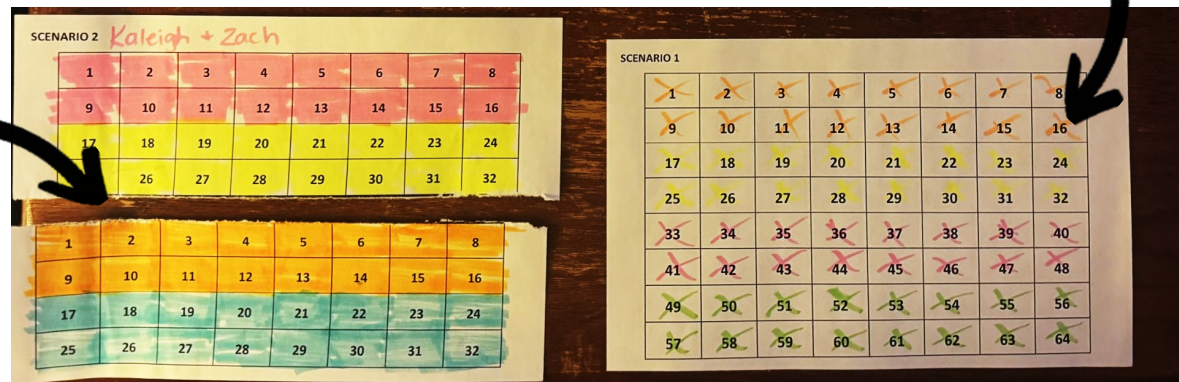
**WRONG!**

**Fill in the squares**



**WRONG!**

**Don't tear the paper!**



# Scenario 1: A Single Processor

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Two people:

1. **The Processor**: colors first stripe (box by box), second stripe (box by box), third stripe (box by box), and then fourth stripe (box by box)
2. **Timer**: Times the Processor

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| P1 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|    | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|    | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
|    | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|    | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
|    | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 |
|    | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 |

Remember: squares must be mostly colored in & you can only color one square at a time!

# Scenario 2: Two Processors Collaborating

Three people:

1. **Processor 1**: Colors red and blue stripes (cell by cell)
2. **Processor 2**: Colors yellow and green stripes (cell by cell)
3. **Timer**: Times them; stop when all cells are colored.

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| P1 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|    | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|    | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| P2 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|    | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|    | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |

Remember: squares must be mostly colored in & you can only color one square at a time!

# Scenario 3: Four Processors Collaborating

Five people:

1. **Processors 1-4**: Each assigned to one color, which they color (cell by cell)
2. **Timer**: Times them; stop when all cells are colored.

|    |   |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|
| P1 | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| P2 | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| P3 | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| P4 | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|    | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |

Remember: squares must be mostly colored in & you can only color one square at a time!

# Scenario 4: Four Processors Collaborating a Different Way (Column Assignments)

Five people:

1. **Processors 1-4**: Each assigned a strip of two columns (first takes left 2 columns, second takes next two columns, etc.) Processors colors these the correct color cell by cell.
2. **Timer**: Times them; stop when all cells are colored.

| P1 |    | P2 |    | P3 |    | P4 |    |
|----|----|----|----|----|----|----|----|
| 1  | 9  | 1  | 9  | 1  | 9  | 1  | 9  |
| 2  | 10 | 2  | 10 | 2  | 10 | 2  | 10 |
| 3  | 11 | 3  | 11 | 3  | 11 | 3  | 11 |
| 4  | 12 | 4  | 12 | 4  | 12 | 4  | 12 |
| 5  | 13 | 5  | 13 | 5  | 13 | 5  | 13 |
| 6  | 14 | 6  | 14 | 6  | 14 | 6  | 14 |
| 7  | 15 | 7  | 15 | 7  | 15 | 7  | 15 |
| 8  | 16 | 8  | 16 | 8  | 16 | 8  | 16 |

Remember: squares must be mostly colored in & you can only color one square at a time!